TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

BE Programme | Year III / Part II

EMBEDDED SYSTEM (CT 655)

QUESTION BANK WITH ANSWERS

Chapter 9: Microcontrollers in Embedded Systems

COURSE SYLLABUS AT A GLANCE

Unit	Topic	Hours	Marks
1	Introduction to Embedded System	3	4
2	Hardware Design Issues	4	8
3	Software Design Issues	6	8
4	Memory	5	8
5	Interfacing	6	8
6	Real-Time Operating System (RTOS)	8	12
7	Control System	3	8
8	IC Technology	3	8
9	Microcontrollers in Embedded Systems	3	8
10	VHDL	4	8
	Total	**45**	**80**

Chapter 9 Topics (3 Hours | 8 Marks):

- 9.1 8051 Microcontroller — Architecture, Block Diagram, Operational Features
- 9.2 8051 Internal Memory Organization (ROM, RAM, SFR, Bit-Addressable Area)
- 9.3 8051 Register Banks (Bank 0–3), PSW, Special Function Registers
- 9.4 8051 Addressing Modes (6 modes with examples)
- 9.5 8051 Timers/Counters — TMOD, TCON, Timer Modes 0–3
- 9.6 8051 Serial Communication — SCON, SBUF, Baud Rate Calculation
- 9.7 8051 Interrupt System — IE, IP Registers, ISR vectors
- 9.8 Assembly Language Programming — Arithmetic, Bit Manipulation, Loops
- 9.9 Seven-Segment Display Interfacing and Counter Programs

Chapter 9: Microcontrollers in Embedded Systems

Q1. What are the important operational features of the 8051 microcontroller? Draw its internal block diagram.	2081 Chaitra (Regular) | 2079 Chaitra (Regular) | 2078 Baishakh (Back) [4+4]

ANSWER

8051 Microcontroller Block Diagram:

```
                    ┌──────────────────────────────────────────┐
                    │            8051 MICROCONTROLLER          │
                    │                                          │
 ┌──────────┐       │  ┌────────┐      ┌──────────────────┐    │
 │ XTAL1/2  │───────┤  │  OSC   │      │   4KB ROM (Code) │    │
 └──────────┘       │  │ /Clock │      │   (0000H-0FFFH)  │    │
                    │  └────────┘      └──────────────────┘    │
                    │      │ Machine                           │
 ┌──────────┐       │      │ Cycles    ┌──────────────────┐    │
 │   RST    │───────┤      │           │128B Internal RAM │    │
 └──────────┘       │  ┌───▼────┐      │(00H-7FH + SFR)   │    │
                    │  │  CPU   │      └──────────────────┘    │
P0 (AD0-AD7)◄───────┤  │ 8-bit  │      ┌──────────────────┐    │
P2 (A8-A15) ◄───────┤  │ ALU    │      │ 2x 16-bit Timers │    │
                    │  │ ACC, B │      │   T0, T1         │    │
P1 (I/O)    ◄───────┤  │ DPTR   │      └──────────────────┘    │
P3 (I/O+alt)◄───────┤  │ PC, SP │      ┌──────────────────┐    │
                    │  └────────┘      │  Full-Duplex UART│    │
INT0, INT1  ►───────┤                  └──────────────────┘    │
TXD, RXD    ◄───────┤                 ┌──────────────────┐     │
                    │                 │ Interrupt Control │     │
                    │                 │ (5 sources, 2 pri)│     │
                    └──────────────────────────────────────────┘
```

Key Operational Features of 8051:

Feature	Specification
CPU	8-bit CISC; 8-bit ALU; 16-bit PC and DPTR
Internal ROM	4 KB program memory (0000H–0FFFH); expandable to 64 KB external
Internal RAM	128 bytes (00H–7FH) + SFR space (80H–FFH)
I/O Ports	4 × 8-bit bidirectional ports: P0, P1, P2, P3 (32 I/O pins total)
Timers/Counters	2 × 16-bit programmable timer/counters (T0, T1); 4 operating modes
Serial Port	Full-duplex UART; programmable baud rate using Timer 1 auto-reload
Interrupts	5 sources: INT0, INT1 (external), TF0, TF1 (timers), RI/TI (serial); 2 priority levels
Instruction Set	111 instructions; mix of 1-, 2-, 3-byte instructions

Clock Speed	Up to 12 MHz standard; 1 machine cycle = 12 clock cycles → 1 MIPS at 12 MHz
Registers	Accumulator A, B register, PSW, SP, DPTR (DPH:DPL), PC
Register Banks	4 banks × 8 registers (R0–R7): Bank 0–3 (addresses 00H–1FH)
Bit Addressing	Bits 00H–7FH in RAM 20H–2FH; individual bits of many SFRs
Power Modes	Normal, Idle (CPU stopped, peripherals run), Power Down (oscillator stopped)
Package	40-pin DIP (standard); also available in smaller SMD packages

P3 Port Alternate Functions:

P3 Pin	Alternate Function	Description
P3.0 (RXD)	Serial input	UART receive data
P3.1 (TXD)	Serial output	UART transmit data
P3.2 (/INT0)	External interrupt 0	Active LOW interrupt input
P3.3 (/INT1)	External interrupt 1	Active LOW interrupt input
P3.4 (T0)	Timer 0 external input	External event counter input
P3.5 (T1)	Timer 1 external input	External event counter input
P3.6 (/WR)	External RAM write strobe	Active LOW write enable
P3.7 (/RD)	External RAM read strobe	Active LOW read enable

Q2. Explain the data memory organization in the 8051 microcontroller with a diagram.	**2079 Chaitra (Regular) | 2078 Poush (Back) | 2079 Jestha (Back) [4+4]**

ANSWER

8051 Internal Data Memory Map (128 bytes, 00H–7FH):

```
Address   Area                  Contents
──────────────────────────────────────────────────────────────

 FFH  ┐
      │  SFR Space             Special Function Registers
 80H  ┘  (Special Fn Regs)    (P0,P1,P2,P3, ACC, B, PSW,
                               SP, DPTR, TMOD, TCON, SCON,
                               SBUF, IE, IP, TH0,TL0,TH1,TL1)
 7FH  ┐
      │  General-Purpose RAM  User variables, stack space
 30H  ┘  (80 bytes)           (Stack grows up from 07H default)

 2FH  ┐
      │  Bit-Addressable RAM  128 individually addressable bits
 20H  ┘  (16 bytes)           Bit addresses: 00H-7FH
                               (e.g., bit 00H = byte 20H bit 0)
```

```
1FH ┐  Register Bank 3     R0-R7  (RS1=1, RS0=1 in PSW)
18H ┘
17H ┐  Register Bank 2     R0-R7  (RS1=1, RS0=0 in PSW)
10H ┘
0FH ┐  Register Bank 1     R0-R7  (RS1=0, RS0=1 in PSW)
08H ┘
07H ┐  Register Bank 0     R0-R7  (RS1=0, RS0=0 — DEFAULT)
00H ┘  (Default Bank)
```

Address Range	Area	Size	Description
00H – 07H	Register Bank 0	8 bytes	Default R0–R7; SP starts at 07H after reset
08H – 0FH	Register Bank 1	8 bytes	R0–R7 selected by PSW RS0=1, RS1=0
10H – 17H	Register Bank 2	8 bytes	R0–R7 selected by PSW RS0=0, RS1=1
18H – 1FH	Register Bank 3	8 bytes	R0–R7 selected by PSW RS0=1, RS1=1
20H – 2FH	Bit-Addressable RAM	16 bytes (128 bits)	Each bit addressable individually (bits 00H–7FH)
30H – 7FH	General-Purpose RAM	80 bytes	User variables and stack area
80H – FFH	SFR Space	128 addresses	Special Function Registers (not all addresses used)

Program Memory (Code Space):

- Internal ROM: 4 KB at addresses 0000H–0FFFH. Contains program code and constant data.
- External ROM: Up to 64 KB accessible via /PSEN (Program Store Enable) signal on P0 (AD0–AD7) and P2 (A8–A15).
- EA pin: When EA=HIGH, 8051 executes from internal ROM (0000H–0FFFH), then automatically fetches from external ROM (1000H–FFFFH). When EA=LOW, all fetches go to external ROM only.
- Reset vector: PC = 0000H after reset — execution starts at address 0000H.
- Interrupt vectors: INT0=0003H, TF0=000BH, INT1=0013H, TF1=001BH, Serial=0023H.

Q3. Explain the register banks in the 8051 microcontroller. Describe the PSW (Program Status Word) register.	**2081 Ashwin (Back) | 2076 Chaitra (Back) [4+4]**

ANSWER

Register Banks in 8051:

The 8051 has four register banks (Bank 0 to Bank 3), each containing 8 general-purpose registers (R0–R7). They share the lower 32 bytes of internal RAM (00H–1FH). Only one bank is active at a time, selected by PSW bits RS1 and RS0.

Bank	RAM Address	RS1 (PSW.4)	RS0 (PSW.3)	When Used
Bank 0	00H – 07H	0	0	Default after reset; most programs use this
Bank 1	08H – 0FH	0	1	ISR can switch to Bank 1 to save context quickly
Bank 2	10H – 17H	1	0	Nested ISR or additional context space
Bank 3	18H – 1FH	1	1	Rarely used; available as general RAM if needed

```
; Switch to Register Bank 1 (for ISR context save):
SETB PSW.3     ; RS0 = 1
CLR  PSW.4     ; RS1 = 0  → selects Bank 1
; Now R0-R7 refers to RAM 08H-0FH

; Return to Bank 0:
CLR PSW.3      ; RS0 = 0
CLR PSW.4      ; RS1 = 0  → Bank 0 active
```

PSW (Program Status Word) — SFR at address D0H:

```
  Bit 7   Bit 6   Bit 5   Bit 4   Bit 3   Bit 2   Bit 1   Bit 0
┌───────┬───────┬───────┬───────┬───────┬───────┬───────┬───────┐
│  CY   │  AC   │  F0   │  RS1  │  RS0  │  OV   │  --   │  P    │
└───────┴───────┴───────┴───────┴───────┴───────┴───────┴───────┘
PSW.7   PSW.6   PSW.5   PSW.4   PSW.3   PSW.2   PSW.1   PSW.0
```

Bit	Name	Function
PSW.7 (CY)	Carry Flag	Set if ADD/SUB produces carry/borrow from bit 7. Also used by rotate and bit instructions.
PSW.6 (AC)	Auxiliary Carry	Set if carry from bit 3 to bit 4 (used for BCD arithmetic with DA A instruction).
PSW.5 (F0)	Flag 0	General-purpose user flag. Set/cleared by programmer.
PSW.4 (RS1)	Register Bank Select 1	Together with RS0 selects active register bank (00–11 for Bank 0–3).
PSW.3 (RS0)	Register Bank Select 0	See RS1 above.
PSW.2 (OV)	Overflow Flag	Set if signed arithmetic overflow occurs (result too large for signed 8-bit).
PSW.1	Reserved	Unused.
PSW.0 (P)	Parity Flag	Hardware-set: 1 if accumulator has odd number of 1s; 0 if even. Used in serial comms.

Q4. Explain the addressing modes of the 8051 microcontroller with examples.	2082 Kartik (Back) | 2079 Ashwin (Back) | 2071 Bhadra (Back) [3+5]

ANSWER

The 8051 supports 6 addressing modes:

1. Immediate Addressing (#data): The operand is an 8-bit or 16-bit constant embedded directly in the instruction.

```
MOV  A,  #25H      ; A ← 0x25 (load constant 0x25 into accumulator)
MOV  DPTR, #2000H  ; DPTR ← 0x2000 (16-bit immediate)
ADD  A,  #0FH      ; A ← A + 15
```

2. Register Addressing: The operand is in one of the 8051's named registers (A, B, R0–R7, DPTR, C).

```
MOV  A, R0         ; A ← R0
ADD  A, R7         ; A ← A + R7
MOV  R2, A         ; R2 ← A
XCH  A, R1         ; Exchange A ↔ R1
```

3. Direct Addressing: An 8-bit address in the instruction specifies the operand location — either internal RAM (00H–7FH) or SFR space (80H–FFH).

```
MOV  A, 30H        ; A ← [RAM address 0x30]
MOV  P1, A         ; [SFR P1 = 0x90] ← A
MOV  30H, #55H     ; [RAM 0x30] ← 0x55
MOV  TCON, #10H    ; direct SFR access
```

4. Register Indirect Addressing (@Ri): Register R0 or R1 holds the address of the operand in internal RAM (or external RAM with MOVX).

```
MOV  R0, #40H      ; R0 ← 0x40 (pointer)
MOV  A, @R0        ; A ← [RAM at address R0] = [0x40]
MOV  @R1, A        ; [RAM at address R1] ← A
MOVX A, @DPTR      ; A ← [External RAM at DPTR] (16-bit pointer)
```

5. Indexed Addressing (@A+Base): The effective address is computed as A + Base register (DPTR or PC). Used to access lookup tables in program memory.

```
MOV  DPTR, #TABLE  ; DPTR points to start of table in code memory
MOV  A, #03H       ; index = 3
MOVC A, @A+DPTR    ; A ← code memory[DPTR + 3] (table lookup)

MOVC A, @A+PC      ; A ← code memory[PC + A] (PC-relative lookup)
```

6. Bit Addressing (bit addr): Individual bits in bit-addressable RAM (20H–2FH, bit addresses 00H–7FH) and many SFRs can be directly accessed.

```
SETB  P1.0         ; Set bit 0 of Port 1 HIGH
CLR   P1.5         ; Clear bit 5 of Port 1 LOW
MOV   C, P3.2      ; Copy P3.2 to Carry flag
JB    P1.0, LABEL  ; Jump if P1.0 = 1
SETB  20H.3        ; Set bit 3 of RAM byte 20H (bit address = 03H)
```

Mode	Instruction Example	Operand Location	Speed
Immediate	MOV A, #25H	In instruction itself	Fastest
Register	MOV A, R0	Named register	Fastest
Direct	MOV A, 30H	Internal RAM or SFR	Fast
Register Indirect	MOV A, @R0	RAM[R0]	Fast (flexible pointer)
Indexed	MOVC A, @A+DPTR	Code memory[DPTR+A]	Slower (code fetch)
Bit	SETB P1.0	Single bit in RAM/SFR	Fast (1 byte instruction)

Q5. Explain the timer/counter operation in the 8051. Describe TMOD and TCON registers. Write a program to generate a 1ms delay using Timer 0 in Mode 1.	**2081 Ashwin (Back) | 2077 Chaitra (Back) | 2075 Bhadra (Back) [4+4]**

ANSWER

8051 Timer/Counter Overview:

The 8051 has two 16-bit timer/counters: Timer 0 (T0) and Timer 1 (T1). Each can operate as a timer (counts internal machine cycles, period = 1µs at 12 MHz) or as an event counter (counts pulses on external T0/T1 pins).

TMOD Register (Timer Mode) — address 89H, not bit-addressable:

```
  Bit 7   Bit 6   Bit 5   Bit 4   Bit 3   Bit 2   Bit 1   Bit 0
┌───────┬───────┬───────┬───────┬───────┬───────┬───────┬───────┐
│ GATE  │  C/T  │  M1   │  M0   │ GATE  │  C/T  │  M1   │  M0   │
└───────┴───────┴───────┴───────┴───────┴───────┴───────┴───────┘
|◄──────────── Timer 1 ────────►|◄──────────── Timer 0 ────────►|

GATE: 1=timer runs only when INTx pin HIGH; 0=timer runs when TRx=1
C/T:  0=Timer (counts machine cycles); 1=Counter (counts external pulses)
M1,M0: Timer mode select:
  00 = Mode 0: 13-bit timer (legacy 8048 compatible)
  01 = Mode 1: 16-bit timer (most common, full 65536 count)
  10 = Mode 2: 8-bit auto-reload timer (fixed period, great for baud rate)
  11 = Mode 3: Split timer (T0 only: two 8-bit timers; T1 stops)
```

TCON Register (Timer Control) — address 88H, bit-addressable:

```
  Bit 7   Bit 6   Bit 5   Bit 4   Bit 3   Bit 2   Bit 1   Bit 0
┌───────┬───────┬───────┬───────┬───────┬───────┬───────┬───────┐
│  TF1  │  TR1  │  TF0  │  TR0  │  IE1  │  IT1  │  IE0  │  IT0  │
└───────┴───────┴───────┴───────┴───────┴───────┴───────┴───────┘


TF1/TF0: Timer 1/0 overflow flag — set by HW when timer overflows; cleared
by SW or RETI
TR1/TR0: Timer 1/0 run control — SET to start, CLR to stop
IE1/IE0: External interrupt 1/0 edge flag
IT1/IT0: INT1/INT0 trigger type: 0=level triggered, 1=edge triggered
```

Timer Calculation (Mode 1, 16-bit):

```
Machine cycle frequency = Crystal / 12 = 11.0592 MHz / 12 = 921,600 Hz
Machine cycle period    = 1 / 921,600 ≈ 1.085 µs

For 1ms delay:
  Counts needed = 1ms / 1.085µs = 921 counts
  Timer initial value = 65536 - 921 = 64615 = FC67H
  TH0 = FCH,  TL0 = 67H

For 12 MHz crystal (simpler math):
  Machine cycle period = 1µs exactly
  Counts for 1ms = 1000
  Timer initial value = 65536 - 1000 = 64536 = FC18H
  TH0 = FCH,  TL0 = 18H
```

Assembly Program: 1ms Delay using Timer 0 Mode 1:

```
        ORG  0000H

MAIN:   ; Generate square wave using 1ms timer delay
        CPL  P1.0          ; Toggle P1.0
        ACALL DELAY_1MS    ; wait 1ms
        SJMP  MAIN         ; repeat forever

; ─── 1ms Delay Subroutine using Timer 0 Mode 1 (12 MHz crystal) ───
DELAY_1MS:
        MOV  TMOD, #01H    ; Timer 0, Mode 1 (16-bit timer)
        MOV  TH0,  #0FCH   ; Load high byte: 64536 = FC18H
        MOV  TL0,  #18H    ; Load low byte
        SETB TR0           ; Start Timer 0
WAIT:   JNB  TF0, WAIT     ; Wait until Timer 0 overflows (TF0=1)
        CLR  TF0           ; Clear overflow flag
        CLR  TR0           ; Stop Timer 0
        RET                ; Return

        END
```

Q6. Explain the interrupt system of the 8051. Describe the IE and IP registers and the interrupt vector table.	2080 Chaitra (Back) | 2078 Chaitra (Back) | 2076 Baisakh (Back) [4+4]

ANSWER

8051 Interrupt System:

The 8051 has 5 interrupt sources with 2 priority levels (high and low). When an interrupt occurs, the CPU completes the current instruction, pushes PC onto the stack, and jumps to the interrupt's fixed vector address.

Interrupt Source	Vector Address	Flag	Description
External INT0 (/P3.2)	0003H	IE0 in TCON	Active LOW or falling edge on INT0 pin
Timer 0 Overflow	000BH	TF0 in TCON	Timer 0 count rolls over from FFFFh to 0000H
External INT1 (/P3.3)	0013H	IE1 in TCON	Active LOW or falling edge on INT1 pin
Timer 1 Overflow	001BH	TF1 in TCON	Timer 1 count rolls over
Serial Port (RI/TI)	0023H	RI or TI in SCON	UART byte received (RI=1) or transmitted (TI=1)

IE Register (Interrupt Enable) — address A8H, bit-addressable:

```
  Bit 7   Bit 6   Bit 5   Bit 4   Bit 3   Bit 2   Bit 1   Bit 0
┌───────┬───────┬───────┬───────┬───────┬───────┬───────┬───────┐
│  EA   │  --   │  --   │  ES   │  ET1  │  EX1  │  ET0  │  EX0  │
└───────┴───────┴───────┴───────┴───────┴───────┴───────┴───────┘


EA  (IE.7): Global interrupt enable — MUST be 1 to allow any interrupt
ES  (IE.4): Enable Serial interrupt
ET1 (IE.3): Enable Timer 1 interrupt
EX1 (IE.2): Enable External interrupt 1
ET0 (IE.1): Enable Timer 0 interrupt
EX0 (IE.0): Enable External interrupt 0

; Enable Timer 0 and External INT0:
MOV IE, #83H    ; 10000011b: EA=1, ET0=1, EX0=1
; or bit by bit:
SETB EA         ; global enable
SETB ET0        ; enable Timer 0 interrupt
SETB EX0        ; enable External INT0
```

IP Register (Interrupt Priority) — address B8H:

```
  Bit 7   Bit 6   Bit 5   Bit 4   Bit 3   Bit 2   Bit 1   Bit 0
┌───────┬───────┬───────┬───────┬───────┬───────┬───────┬───────┐
│  --   │  --   │  --   │  PS   │  PT1  │  PX1  │  PT0  │  PX0  │
└───────┴───────┴───────┴───────┴───────┴───────┴───────┴───────┘
```

```
Each bit: 0 = low priority (default); 1 = high priority
High priority ISR can interrupt a low priority ISR (nested interrupts)

; Set Timer 0 to HIGH priority (can interrupt other ISRs):
SETB PT0        ; Timer 0 = high priority

; Default: all low priority, priority order by vector address:
; INT0 > TF0 > INT1 > TF1 > Serial (when same priority level)
```

Interrupt Service Routine (ISR) Example:

```
        ORG  0000H
        LJMP MAIN           ; Jump over interrupt vectors

        ORG  000BH          ; Timer 0 interrupt vector
        LJMP TIMER0_ISR     ; Jump to Timer 0 ISR

        ORG  0100H          ; Main program
MAIN:   MOV  TMOD, #01H    ; Timer 0, Mode 1
        MOV  TH0,  #0FCH   ; 1ms reload value
        MOV  TL0,  #18H
        MOV  IE,   #82H    ; EA=1, ET0=1
        SETB TR0            ; Start Timer 0
        SJMP $              ; Main loop (CPU free for other tasks)

TIMER0_ISR:                 ; Runs every 1ms
        MOV  TH0, #0FCH    ; Reload timer for next 1ms
        MOV  TL0, #18H
        CPL  P1.0           ; Toggle LED every 1ms
        RETI                ; Return from interrupt, restore PC
```

Q7. Write an 8051 assembly program to read an 8-bit temperature in Celsius from Port 1 and output the Fahrenheit equivalent on Port 2. (F = C × 9/5 + 32)	**2082 Kartik (Back) | 2078 Poush (Back) [3+5]**

ANSWER

Algorithm:

1. Read Celsius value from Port 1 into Accumulator A.
2. Multiply A by 9 using MUL AB (B=9). Result in B:A (16-bit).
3. Divide B:A by 5 using DIV AB. Quotient in A, remainder in B.
4. Add 32 to A.
5. Output A to Port 2.
6. Repeat continuously.

```
;————————————————————————————————————————————————————————————
; 8051 Assembly: Celsius to Fahrenheit Converter
; Input:  Port 1 (P1) = Celsius temperature (0-99°C, 8-bit)
; Output: Port 2 (P2) = Fahrenheit temperature
; Formula: F = (C × 9) / 5 + 32
; Note: For C ≤ 28, C×9 ≤ 252 (fits in 8 bits, no overflow in A)
;       For C > 28, use 16-bit result in B:A before dividing
;————————————————————————————————————————————————————————————
        ORG   0000H

START:  MOV   A,  P1       ; Step 1: Read Celsius from Port 1

        MOV   B,  #09H     ; Step 2: B = 9
        MUL   AB           ; A × B → 16-bit result: B (high) : A (low)
                           ; e.g., C=100: 100×9=900=0384H → B=03H, A=84H

        ; Handle 16-bit result for large Celsius values:
        ; If B > 0 (result > 255), need to divide 16-bit number by 5
        ; Simple approach valid for C ≤ 28°C (result ≤ 252 fits in A):

        MOV   B,  #05H     ; Step 3: B = 5
        DIV   AB           ; A = A / 5 (quotient), B = remainder

        ADD   A,  #32      ; Step 4: A = A + 32
        MOV   P2, A        ; Step 5: Output Fahrenheit to Port 2

        SJMP  START        ; Repeat continuously
        END

; Example trace for C = 25°C:
;   A = 25, B = 9  → MUL: A = 225 (0xE1), B = 0
;   A = 225, B = 5 → DIV: A = 45 (quotient), B = 0
;   A = 45 + 32 = 77 → P2 = 77 = 0x4D  ✓ (25°C = 77°F)
```

Q8. Write 8051 assembly programs to: (a) Count number of 0s in an 8-bit data at RAM address 40H and store result at 41H. (b) Count number of 1s in data at 75H and store at 45H.

2079 Chaitra (Regular) | 2079 Ashwin (Back) | 2078 Poush (Back) [4+4]

ANSWER

(a) Count number of 0s in data at RAM address 40H:

```
;————————————————————————————————————————————————————————————
; Count 0s in 8-bit data at RAM[40H], store result at RAM[41H]
; Method: Rotate right through carry 8 times.
;         If carry = 0 after rotate → that bit was 0 → increment count
;————————————————————————————————————————————————————————————
        ORG   0000H
```

```
START:  MOV   A,   40H    ; Load data from RAM address 40H
        MOV   R0,  #08H   ; R0 = bit counter (8 bits to check)
        MOV   R1,  #00H   ; R1 = zero count = 0

LOOP:   RRC   A         ; Rotate right through Carry: LSB → Carry, MSB ← 0
        JC    SKIP        ; If Carry = 1, the bit was 1 → skip (don't
count)
        INC   R1          ; Carry = 0 → bit was 0 → increment zero count
SKIP:   DJNZ  R0, LOOP    ; Decrement bit counter; loop if not zero

        MOV   41H, R1     ; Store count of 0s at RAM address 41H
        SJMP  $           ; Halt (infinite loop)
        END

; Example: Data at 40H = 0xA5 = 10100101b
;   Bits: 1,0,1,0,0,1,0,1 → three 0s (bits 1,3,4) → R1=3?
;   Count: positions 1,3,4 are 0 → 3 zeros... wait:
;   10100101: 1s at bits 7,5,2,0 → four 1s; 0s at bits 6,4,3,1 → four 0s
;   Result: 41H = 04H
```

(b) Count number of 1s in data at RAM address 75H:

```
;————————————————————————————————————————————————————————————
; Count 1s in 8-bit data at RAM[75H], store result at RAM[45H]
; Method: Rotate right through carry 8 times.
;         If carry = 1 → that bit was 1 → increment count
;————————————————————————————————————————————————————————————
        ORG   0000H

START:  MOV   A,   75H    ; Load data from RAM address 75H
        MOV   R0,  #08H   ; R0 = bit counter = 8
        MOV   R1,  #00H   ; R1 = one-count = 0

LOOP:   RRC   A           ; Rotate right: LSB → Carry
        JNC   SKIP        ; If Carry = 0, bit was 0 → skip
        INC   R1          ; Carry = 1 → bit was 1 → increment count
SKIP:   DJNZ  R0, LOOP    ; Loop for all 8 bits

        MOV   45H, R1     ; Store count of 1s at RAM address 45H
        SJMP  $           ; Halt
        END

; Example: Data = 0xB3 = 10110011b
;   1s at bits 7,5,4,1,0 → five 1s → 45H = 05H
```

Q9. (a) Write an assembly program to alternately blink 8 LEDs on Port 2. (b) Write a program to generate a 50% duty cycle square wave on P1.5.

2081 Chaitra (Regular) | 2081 Ashwin (Back) | 2075 Baisakh (Back) [4+4]

ANSWER

(a) Blink 8 LEDs on Port 2 (alternate ON/OFF with ~500ms delay):

```
;──────────────────────────────────────────────────────────────
; Blink all 8 LEDs on Port 2 (active HIGH: 1=LED ON, 0=LED OFF)
; Delay ~500ms using nested loop at 11.0592 MHz
;──────────────────────────────────────────────────────────────
        ORG   0000H

START:  MOV   P2, #0FFH   ; Turn ON all 8 LEDs (P2 = 11111111b)
        ACALL DELAY        ; Wait ~500ms
        MOV   P2, #00H    ; Turn OFF all 8 LEDs (P2 = 00000000b)
        ACALL DELAY        ; Wait ~500ms
        SJMP  START        ; Repeat forever

;─── Delay Subroutine: ~500ms at 11.0592 MHz ───
; Inner loop: DJNZ R1 = 2 machine cycles × 250 = 500 cycles ≈ 543µs
; Outer loop: 250 iterations × 543µs ≈ 135ms... ×4 nested levels ≈ 500ms
DELAY:  MOV   R0, #10     ; Outer outer loop count
D1:     MOV   R1, #200    ; Outer loop count
D2:     MOV   R2, #250    ; Inner loop count
D3:     DJNZ  R2, D3      ; Inner loop: 250 × 2 cycles = 500 cycles
        DJNZ  R1, D2      ; Middle loop
        DJNZ  R0, D1      ; Outer loop
        RET                ; Return to caller
        END
```

(b) 50% Duty Cycle Square Wave on P1.5:

```
;──────────────────────────────────────────────────────────────
; Generate 50% duty cycle square wave on P1.5
; CPL instruction toggles pin each iteration → perfect 50% duty cycle
; Frequency determined by DELAY subroutine (half-period = DELAY time)
;──────────────────────────────────────────────────────────────
        ORG   0000H

START:  CPL   P1.5        ; Toggle P1.5 (complement: 0→1 or 1→0)
        ACALL HALF_PERIOD  ; Wait for half-period
        SJMP  START        ; Repeat: gives 50% duty cycle

;─── Half-Period Delay (~1ms for ~500Hz square wave) ───
HALF_PERIOD:
        MOV   TMOD, #01H  ; Timer 0, Mode 1 (16-bit)
```

```
        MOV   TH0,  #0FCH ; 1ms at 12 MHz: 65536-1000=64536=FC18H
        MOV   TL0,  #18H
        SETB  TR0          ; Start timer
WAIT:   JNB   TF0, WAIT   ; Wait for overflow (TF0 goes HIGH)
        CLR   TF0          ; Clear overflow flag
        CLR   TR0          ; Stop timer
        RET
        END

; Result: P1.5 toggles every 1ms → square wave period = 2ms → f = 500 Hz
; Duty cycle = exactly 50% because both HIGH and LOW last exactly 1ms
```

Q10. Explain seven-segment display with its configurations. Design a 2-digit hex counter using Port 1 (LSB) and Port 0 (MSB) with 2ms delay using Timer 0 Mode 1.

2080 Ashwin (Back) | 2077 Chaitra (Back) | 2075 Bhadra (Back) [2+6]

ANSWER

Seven-Segment Display:

```
Segment Layout:      Segment to Port Pin Mapping:
   — a —            Port bit  7 6 5 4 3 2 1 0
  |     |           Segment  dp g f e d c b a
  f     b
  |     |             Digit patterns (Common Cathode, active HIGH):
   — g —              '0' = 0011 1111 = 3FH  (a,b,c,d,e,f ON)
  |     |             '1' = 0000 0110 = 06H  (b,c ON)
  e     c             '2' = 0101 1011 = 5BH
  |     |             '3' = 0100 1111 = 4FH
   — d —  .dp         '4' = 0110 0110 = 66H
                      '5' = 0110 1101 = 6DH
                      '6' = 0111 1101 = 7DH
Two configurations:   '7' = 0000 0111 = 07H
                      '8' = 0111 1111 = 7FH
Common Anode (CA):    '9' = 0110 1111 = 6FH
  All anodes → VCC    'A' = 0111 0111 = 77H
  Segment ON = LOW    'B' = 0111 1100 = 7CH
                      'C' = 0011 1001 = 39H
Common Cathode (CC):  'D' = 0101 1110 = 5EH
  All cathodes → GND  'E' = 0111 1001 = 79H
  Segment ON = HIGH   'F' = 0111 0001 = 71H
```

```
;————————————————————————————————————————————————————————————
; 2-Digit Hex Counter: P1 = LSB digit (00-0F), P0 = MSB digit (00-F0)
; 2ms delay between counts using Timer 0 Mode 1 @ 11.0592 MHz
; Common Cathode 7-segment display (active HIGH segment codes)
;————————————————————————————————————————————————————————————
        ORG   0000H
```

```
        LJMP  START

;——— 7-segment lookup table for digits 0-F ———
SEG_TABLE:
        DB  3FH,06H,5BH,4FH,66H,6DH,7DH,07H   ; 0,1,2,3,4,5,6,7
        DB  7FH,6FH,77H,7CH,39H,5EH,79H,71H   ; 8,9,A,B,C,D,E,F

        ORG   0100H
START:  MOV   R2, #00H     ; R2 = MSB digit (0-F)

LOOP1:  MOV   R3, #00H     ; R3 = LSB digit (0-F)

LOOP2:  MOV   DPTR, #SEG_TABLE

        ; — Display LSB digit on Port 1 —
        MOV   A,  R3
        MOVC  A,  @A+DPTR  ; Lookup segment code for LSB digit
        MOV   P1, A         ; Output to Port 1

        ; — Display MSB digit on Port 0 —
        MOV   A,  R2
        MOVC  A,  @A+DPTR  ; Lookup segment code for MSB digit
        MOV   P0, A         ; Output to Port 0

        ; — 2ms delay using Timer 0 Mode 1 —
        ; @ 11.0592 MHz: MC period = 1.085µs
        ; 2ms / 1.085µs = 1843 counts
        ; 65536 - 1843 = 63693 = F8CDH
        MOV   TMOD, #01H   ; T0 Mode 1 (16-bit)
        MOV   TH0,  #0F8H  ; High byte
        MOV   TL0,  #0CDH  ; Low byte
        SETB  TR0           ; Start Timer 0
TWAIT:  JNB   TF0, TWAIT   ; Wait for overflow
        CLR   TF0           ; Clear flag
        CLR   TR0           ; Stop timer

        INC   R3            ; Next LSB value
        CJNE  R3, #10H, LOOP2  ; If LSB < 16, repeat inner loop

        INC   R2            ; Next MSB value
        CJNE  R2, #10H, LOOP1  ; If MSB < 16, repeat outer loop

        SJMP  START         ; Restart from 00 after FF
        END
```

Q11. Explain the serial communication of the 8051. Describe SCON and SBUF registers. Write a program to transmit the character 'A' continuously at 9600 baud.

2080 Chaitra (Back) | 2077 Baisakh (Back) [4+4]

ANSWER

8051 Serial Communication:

The 8051 has a built-in full-duplex UART (Universal Asynchronous Receiver/Transmitter). It uses P3.1 (TXD) for transmit and P3.0 (RXD) for receive. Baud rate is typically generated using Timer 1 in Mode 2 (8-bit auto-reload).

SCON Register (Serial Control) — address 98H, bit-addressable:

```
  Bit 7   Bit 6   Bit 5   Bit 4   Bit 3   Bit 2   Bit 1   Bit 0
┌───────┬───────┬───────┬───────┬───────┬───────┬───────┬───────┐
│  SM0  │  SM1  │  SM2  │  REN  │  TB8  │  RB8  │  TI   │  RI   │
└───────┴───────┴───────┴───────┴───────┴───────┴───────┴───────┘


SM0, SM1: Serial mode select:
  00 = Mode 0: Shift register (fixed 1/12 of crystal frequency)
  01 = Mode 1: 8-bit UART, variable baud (most common) ← use this
  10 = Mode 2: 9-bit UART, fixed baud (1/32 or 1/64 of crystal)
  11 = Mode 3: 9-bit UART, variable baud
SM2: Multiprocessor communication enable
REN: Receive Enable — must be SET (1) to enable reception
TB8: 9th bit to transmit (Mode 2/3 only)
RB8: 9th bit received (Mode 2/3 only)
TI:  Transmit Interrupt flag — SET by HW when byte transmitted; cleared by
SW
RI:  Receive Interrupt flag  — SET by HW when byte received; cleared by SW
```

Baud Rate Calculation (Timer 1, Mode 2):

```
Baud Rate = (Crystal / 32) / (256 - TH1)    [when SMOD=0 in PCON]
          = (Crystal / 12 / 32) / (256 - TH1)  [machine cycles]


For 9600 baud with 11.0592 MHz crystal:
  9600 = 11059200 / (32 × 12 × (256 - TH1))
  256 - TH1 = 11059200 / (32 × 12 × 9600) = 3
  TH1 = 256 - 3 = 253 = FDH  ← load into TH1 (auto-reload in Mode 2)

Common baud rate reload values (11.0592 MHz, SMOD=0):
  9600 baud  → TH1 = FDH
  4800 baud  → TH1 = FAH
  2400 baud  → TH1 = F4H
  1200 baud  → TH1 = E8H


Note: 11.0592 MHz is chosen specifically because it gives exact baud rates
      (no rounding error). 12 MHz crystal gives ~0.16% error at 9600 baud.
```

Assembly Program: Transmit 'A' (0x41) continuously at 9600 baud:

```
        ORG   0000H

; ─── Initialize Serial Port at 9600 baud ───
INIT:   MOV   TMOD, #20H   ; Timer 1, Mode 2 (8-bit auto-reload)
        MOV   TH1,  #0FDH  ; Reload = FDH for 9600 baud @ 11.0592MHz
        MOV   TL1,  #0FDH  ; Initial value same as reload
        SETB  TR1           ; Start Timer 1 (baud rate generator)
        MOV   SCON, #50H   ; 01010000b: Mode 1 (8-bit UART), REN=1
                           ; SM0=0,SM1=1,SM2=0,REN=1,TB8=0,RB8=0,TI=0,RI=0

; ─── Transmit 'A' = 0x41 continuously ───
MAIN:   MOV   SBUF, #41H   ; Load 'A' into Serial Buffer → starts
transmission
WAIT:   JNB   TI, WAIT     ; Wait until TI=1 (transmission complete)
        CLR   TI            ; Clear TI flag (ready for next byte)
        SJMP  MAIN          ; Transmit again
        END
```

Summary: Topics Covered in Chapter 9

S.N.	Topic	Exam(s) Asked	Marks
1	8051 Architecture — Block Diagram + Operational Features Table	2081 Chaitra, 2079 Chaitra, 2078 Baishakh	4+4
2	P3 Alternate Functions Table	2081 Chaitra, 2079 Chaitra	2
3	Internal Data Memory Organization (full map 00H–FFH)	2079 Chaitra, 2078 Poush, 2079 Jestha	4+4
4	Program Memory (ROM, EA pin, interrupt vectors)	2079 Chaitra, 2078 Poush	4
5	Register Banks (Bank 0–3) + RAM addresses + PSW RS bits	2081 Ashwin, 2076 Chaitra	4+4
6	PSW Register — all 8 bits explained	2081 Ashwin, 2076 Chaitra	4
7	All 6 Addressing Modes with code examples + comparison table	2082 Kartik, 2079 Ashwin, 2071 Bhadra (annual)	3+5
8	TMOD Register — all bits, all 4 modes	2081 Ashwin, 2077 Chaitra, 2075 Bhadra	4
9	TCON Register — TF1/TR1/TF0/TR0 bits	2081 Ashwin, 2077 Chaitra	4
10	Timer Calculation (Mode 1, 16-bit) + 1ms delay program	2081 Ashwin, 2077 Chaitra, 2075 Bhadra	4+4
11	Interrupt System — 5 sources + vector table	2080 Chaitra, 2078 Chaitra, 2076 Baisakh	4
12	IE and IP Registers — all bits with code	2080 Chaitra, 2078 Chaitra, 2076 Baisakh	4
13	Celsius to Fahrenheit Program	2082 Kartik, 2078 Poush	3+5
14	Count 0s in 8-bit data (RRC loop method)	2079 Chaitra (annual)	4
15	Count 1s in 8-bit data (RRC loop method)	2079 Ashwin (annual)	4
16	Blink 8 LEDs on Port 2 (with delay subroutine)	2081 Chaitra, 2072 Magh, 2075 Baisakh (annual)	4
17	50% Duty Cycle Square Wave on P1.5 (CPL + timer)	2081 Ashwin, 2078 Chaitra	4
18	7-Segment Display — CC/CA configs + segment codes table	2080 Ashwin, 2077 Chaitra, 2075 Bhadra	2
19	2-Digit Hex Counter (P1=LSB, P0=MSB, Timer 0 Mode 1, 2ms)	2080 Ashwin, 2077 Chaitra	6
20	SCON Register — all modes, all bits	2080 Chaitra, 2077 Baisakh	4
21	Baud Rate Calculation (Timer 1 Mode 2) + common values table	2080 Chaitra, 2077 Baisakh	4
22	Serial Transmit Program — send 'A' at 9600 baud	2080 Chaitra, 2077 Baisakh	4